Loan Default Prediction using Random Forest — Technical Sheet

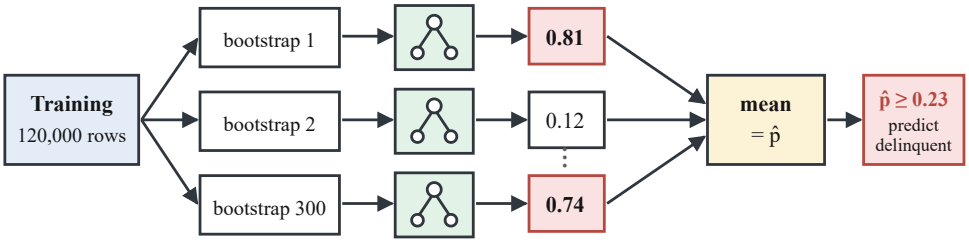
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1 · Problem

Supervised binary classification. From 10 numeric borrower attributes, estimate P(serious delinquency within 2 years), then threshold it to a class. Not regression (the output is a category) and not clustering (the groups come from labels).

150,000	10	6.68%
borrowers	features	positive

2 · Random Forest — how a prediction is made



- **Bag.** 300 bootstrap samples, drawn with replacement.
- **Split.** At each node search a *random feature subset*; take the split that most reduces Gini.
- **Stop.** Leaf must hold ≥ 20 samples.
- **Predict.** Each tree gives its leaf's positive proportion; the forest **averages** them — not a majority vote.

Gini impurity

$$G = 1 - \Sigma\, p_k^2$$

Lower = purer node. The split maximises the weighted drop in G.

3 · Evaluation metrics — from TP / FP / TN / FN

Accuracy

$$\frac{TP + TN}{N}$$

Misleading when one class is rare

Precision

$$\frac{TP}{TP + FP}$$

Of those flagged, how many were right

Recall (TPR)

$$\frac{TP}{TP + FN}$$

Of the truly delinquent, how many found

F1

$$2 \cdot \frac{P \cdot R}{P + R}$$

Harmonic mean — the weaker one dominates

FPR

$$\frac{FP}{FP + TN}$$

Divides by 27,995 negatives → flattering

ROC-AUC

$$P(s^+ > s^-)$$

Area under TPR–FPR. Chance = 0.5

PR-AUC

$$\int P\, dR$$

Baseline = positive rate = 0.0668

Why PR-AUC here

FPR's huge denominator hides false alarms. Precision divides only by what was flagged, so it stays honest.

4 · Results — held-out test set (30,000 borrowers, scored once)

	pred. 0	pred. 1
actual 0	26,621 TN	1,374 FP · false alarm
actual 1	1,029 FN · missed	976 TP · caught

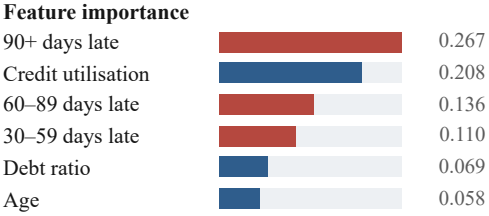
Caught 976 of 2,005 delinquents at cut-off 0.2302.

0.8654	0.4014		
ROC-AUC	PR-AUC		
0.4868	0.4153	0.4482	
Recall	Precision	F1	

Accuracy 0.9199 — below the 93.3% of always answering "no". That is the point.

5-fold CV: ROC-AUC 0.8632 ± 0.0053. Gap train−test: +0.0675 ROC, +0.1218 PR.

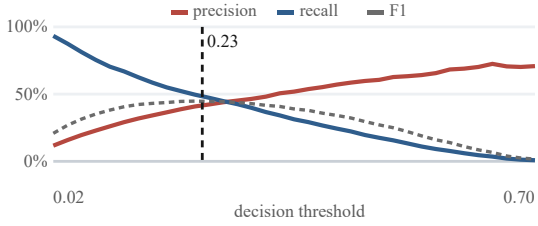
Feature importance



Payment history (red) = 51%. Describes splits, not causation.

5 · Class imbalance and the decision threshold

- Only **6.68%** positive. Three remedies: resampling, class weighting, moving the threshold.
- **Class weighting tested, rejected** — recall 0.1556 → 0.7197 but PR-AUC 0.3820 → 0.3776: it slides *along* the curve, it does not lift it.
- **Threshold chosen on validation** by maximising F1 → **0.2302**, not 0.50. Validation F1 0.4396 → test 0.4482, so it generalised.
- **Protocol.** Stratified 60/20/20; statistics fitted on train only; 5 configurations compared on validation; test touched once.



Limitations. US data from ~2011 with no date column, so no out-of-time validation — present-day performance unknown. Probabilities are **uncalibrated**: 0.30 is a ranking score, not "30 in 100 default". No protected attributes, so fairness cannot be tested — absence of measured bias is not evidence of fairness. Maximising F1 weights a miss and a false alarm equally; lending economics do not. **Educational prototype — not for real credit decisions.**